

## AI for Research & Science

Three use-cases



Presentation at the NEO-AMELEAS Workshop

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# Outline

## I) AI for Skeptical People

- Creating Vector Images with TikZ
- Saving Time with AI: Creating a Counterexample

## II) AI for Enthusiasts

- GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)

# 01

## AI for Skeptical People



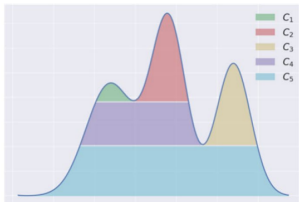
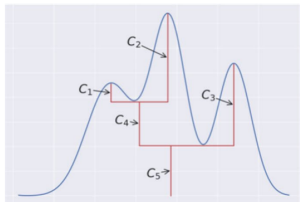


## Enhancing Visual Quality: Fast Vector Graphics with AI

- ▶ **Fast & Professional Figures:** AI allows you to generate high-quality illustrations quickly without wasting time.
- ▶ **What is TikZ?:** A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- ▶ **Full Control:** Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.



## Example: Target Raster Image

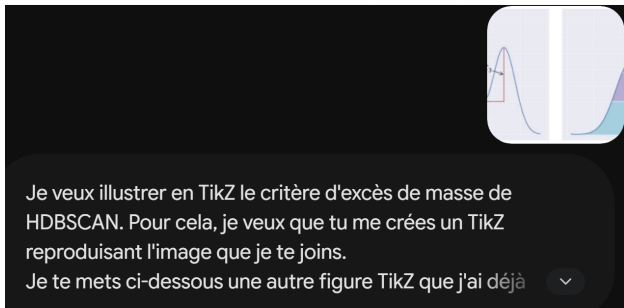


### The Challenge:

- ▶ A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- ▶ Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.



## Example: Prompting the AI

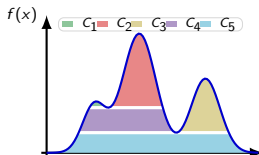
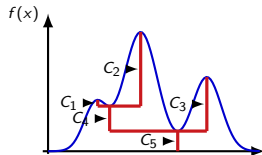


### Human-AI Collaboration:

- ▶ Simply upload the target image and ask the AI to reproduce it in TikZ.
- ▶ Saves hours of calculating coordinates and writing syntax manually.



## Example: The Native TikZ Result



### Why it works:

- ▶ Native vector rendering means infinite scaling and crispness.
- ▶ Complete control over font size and styling.



## Saving Time with AI: Creating a Counterexample





# 02

## AI for Enthusiasts





## GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)



Thank you!